

Henry Demarest

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EDUCATION

Stanford University

Master's Degree in Aerospace Engineering

- **GPA:** 3.96

Palo Alto, CA

Sept. 2025 – June, 2027

Yale University

BS in Mechanical Engineering (ABET), BS in Computer Science

- **GPA:** 3.98
- **Awards:** Summa Cum Laude, Phi Beta Kappa, Tau Beta Pi, L.C. Lichty & E.O. Watters Prize in Mechanical Engineering, Matthew Isakowitz Fellowship, Robert E. Lewis Award, Wesley M. Oler Prize
- **Non-Technical Activities:** First-Year Counselor, Vice Chair of Yale ASME, MechE Peer Mentor, Engineering Tour Guide, Yale Club Running, DPods Orchestra, Backpacking Trip Leader, Intramural Secretary

New Haven, CT

Aug. 2021 – May 2025

Irvington High School

High School Degree

- **Weighted GPA:** 4.58, **SAT:** 1600
- **Awards:** Valedictorian (2021), National Merit Semifinalist (2021), U.S. Presidential Scholar Semi-Finalist (2021)

Irvington, NY

Sept. 2017 – June, 2021

WORK EXPERIENCE

Relativity Space

GNC Engineering Intern

- Fully owned the development, tuning, and analysis of a novel flight algorithm to be used on TerranR's ascent
- Maintained and improved a monte carlo simulation case related to pad drift on launch

Long Beach, CA

June 2026 - Sept 2026

GNC Simulation Engineering Intern

- Developed an improved architecture for modeling rocket engine dynamics during flight
- Integrated an improved aerodynamics model for stage interactions during separation into high fidelity simulation

May 2025 - Aug 2025

GNC Performance Engineering Intern

- Led key trade studies focused on optimizing vehicle trajectory via different loads-informed throttle controllers
- Implemented automated feature to improve link between monte carlo data and trajectory optimization framework

May 2024 - Aug 2024

Astrobotic Technologies

Lunar Surface Systems - Mechanical Engineering Intern

- Designed test stands to subject components (solar arrays and dust covers) to cryogenic/vacuum environments
- Iterated upon the design of a spacerless-MLI thermal enclosure to enable spacecraft survival of the lunar night

Pittsburgh, PA

May 2023 - Aug 2023

PROJECTS

Yale Undergraduate Aerospace Association

Executive Board - President

- Led the executive board, and engaged with sponsors, administrators, aerospace companies, and other stakeholders
- Hosted inter-college conferences, industry recruitment events, volunteering opportunities, and technical workshops

New Haven, CT

May 2023 - May 2024

Project Liquid: Chief Engineer (2023-24), Co-President (2022-23), Propulsion Lead (2021-22) *Sept 2021 - June 2025*

- Led the development of a regeneratively-cooled liquid rocket using nitrous oxide and ethane as propellants
- Managed the design, manufacturing, and testing of a custom spark ignitor
- Designed a coaxial shear injector using a stacked manifold design as Propulsion Lead

CubeSat Team – Attitude Determination and Control System (ADCS) Lead (2022-23)

Sept 2021 - June 2023

- Led the ADCS Subteam to prepare CubeSat for launch as part of NASA's CubeSat Launch Initiative
- Designed and implemented an algorithm to control CubeSat's active stabilization efforts using magnetorquers

Heat Shield Tile Impact Testing

Mechanical Engineering Senior Design Student - Nanoprobe Group

New Haven, CT

June 2024 - May 2025

- Worked with NASA Ames Research Center to conduct impact testing on spacecraft heat shield tiles
- Designed, built, and tested a custom impact hammer, spring-powered silder, and penumatic pellet launcher

Quadcopter Ergodic Controller Implementation

New Haven, CT

Computer Science Senior Design Student - Intelligent Autonomy Lab

June 2024 - May 2025

- Wrote code to implement a decentralized ergodic search algorithm in a CrazyFlie drone's embedded controller
- Conducted preliminary testing in the lab's drone cage to validate drone's operation

TEACHING

Yale University - CS223 (Data Structures & Programming Techniques)

New Haven, CT

Undergraduate Learning Assistant

Sept 2022 - Dec 2023

- Attended weekly teaching staff meetings for CPSC 223 (Data Structures & Programming Techniques) and provided input on the course
- Held office hours and teach students topics ranging from specific programming syntax to general problem solving strategies

Yale University - ENAS 772 (Introduction to Embedded Robotic Systems)

New Haven, CT

Undergraduate Learning Assistant

Jan 2025 - May 2025

- Hold office hours with students to go over course material for Introduction to Embedded Robotic Systems
- Grade homework assignments and providing useful feedback for students

TEACHING

Stanford NavLab

Palo Alto, CA

Master's Student Researcher

Sept 2025 - Present

- Developing SLAM (simultaneous localization and mapping) benchmarking capabilities for the lab
- Working on team to improve autonomous lunar navigation systems

Yale Nanoprobe Group

New Haven, CT

Undergraduate Student Researcher

Jan 2023 - May 2025

- Conducted research on the microstructural effects of aging MP35N, a nickel-chromium superalloy
- Using nanoindentation, scanning electron microscopy, x-ray diffraction, and other advanced analysis techniques

Yale Intelligent Autonomy Lab

New Haven, CT

Undergraduate Student Researcher

Sept 2024 - May 2025

- Developing a decentralized, multi-agent ergodic search implementation to run on Crazyflie nano-quadcopters
- Working with C firmware to efficiently calculate ergodic trajectories onboard drones

Cube Satellite Detumbling Research Project

New Haven, CT

Undergraduate Student Researcher

May 2022 - Aug 2022

- Designed and simulated detumbling algorithm for a 2U CubeSat with limited sensors and actuators
- Used TRIAD algorithm for attitude determination and B-dot controllers paired with magnetorquers for control

Machine Learning Research Project with Mentors at IBM

Yorktown, NY

High School Student Researcher

January 2019 - May 2021

- Assessed the performance of reinforcement learning algorithms with a bipedal walking task using Python code
- Earned 1st Place at NYS Science and Engineering Fair (NYSSEF) and attended Regeneron ISEF competition

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TECHNICAL SKILLS

Programming Languages: C, C++, D, Java, Python, MATLAB, Julia, Rust

CAD: Solidworks, Autodesk Inventor, Fusion360, OnShape

Manufacturing: Lathe, Mill, CNC, 3D-Printer, Hand Tools

Software: Project Management (Confluence/Jira), Collaboration (GitLab, GitHub, Solidworks PDM), Microsoft Office